Research Survey on Human-Centric Narrative Urban Sensing

— Originality Assessment of the HOHOHO Model
and Prospects for Cross-Sector Collaboration —

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HOHOHO Project / MENTA Mentoring

Multi-LLM Cross-Validation: Claude (Opus 4.6) + Gemini DeepResearch

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Abstract

This research survey evaluates the academic originality of the HOHOHO model, a framework that leverages citizen-generated narrative, emotional, and photographic data collected during walking for urban infrastructure management. Employing a multi-LLM cross-validation methodology using Claude (Opus 4.6) and Gemini DeepResearch, we conducted a comprehensive review of peer-reviewed publications and commercial products across Western academic databases including Google Scholar, Semantic Scholar, IEEE, ACM Digital Library, Springer, and Elsevier.

Our principal finding is that while each constituent element of HOHOHO—participatory sensing, emotional cartography, walking survey methods, NLP-based urban analysis, volunteered geographic information, and citizen science—is rooted in an established research domain, no end-to-end model integrating "walking × citizen narrative/emotion × AI/LLM processing × urban infrastructure application" exists in the peer-reviewed literature or as a commercial product. We identify six structural gaps in existing research and discuss how cross-sector collaboration between field operators, AI technology providers, universities, and municipal governments could address these gaps.

Keywords: *participatory sensing, emotional cartography, narrative urban data, walkability, citizen science, NLP, infrastructure management, LLM cross-validation*

1. Introduction

1.1 Background and Problem Statement

Japan's urban infrastructure faces a critical inflection point. Of the nation's approximately 730,000 road bridges, 63% will exceed 50 years of age by 2033[1]. Meanwhile, roughly 25% of municipalities have zero civil engineering technical staff, and a majority of inspection work is outsourced to external contractors. The National Resilience Implementation Mid-Term Plan (FY2026–2030) allocates over 20 trillion yen, yet existing vehicle-mounted AI inspection systems and IoT sensors have a structural limitation: they cannot detect anomalies on school routes, pedestrian walkways, or narrow residential streets.

The HOHOHO project addresses this challenge by positioning citizen walking as the core mechanism for data collection. A proof-of-concept (PoC) conducted in Fukui City demonstrated that a single walker operating one hour per day for one month could detect 367 infrastructure anomalies, confirming that municipal administrators prioritize "decision-actionability" (i.e., prioritized triage) over raw detection volume.

1.2 Research Questions

This survey addresses six research questions: (1) Do peer-reviewed studies exist where citizens contribute walking experiences or emotions as urban data? (2) Does this fall under participatory sensing, human sensing, or citizen sensing? (3) Are there studies incorporating narrative data into urban analysis? (4) What are the representative smart city projects involving citizen-participatory data collection? (5) Do actual deployed products or services exist? (6) What is the research trend for "human-centered urban sensing" distinct from Google Street View or IoT approaches?

1.3 Methodology: Multi-LLM Cross-Validation

This survey employed a novel "cross-validation" approach using multiple LLMs. Claude (Opus 4.6) served as the primary survey engine, searching academic databases. In parallel, Gemini DeepResearch was given an identical prompt (a YAML-formatted research question specification) to conduct an independent survey.

Critically, Claude performed a systematic hallucination check on Gemini's output, verifying URL existence, cross-checking factual claims, confirming citation-content correspondence, and validating dates and proper nouns. This revealed that approximately 15–20% of Gemini's 54 citations pointed to sources unrelated to the claims they supported. The present paper adopts only findings where both LLMs converged, and explicitly flags points of divergence.

[1] Ministry of Land, Infrastructure, Transport and Tourism (MLIT), Road Maintenance Annual Report FY2023. Approx. 25% of municipalities have zero civil engineering technical staff.

2. Overview of Related Research Fields

HOHOHO intersects six established research domains. However, it is precisely the intersection of these fields that remains unexplored, and understanding this is key to positioning the model.

2.1 Participatory Sensing

Participatory sensing was conceptualized in 2006 by Burke, Estrin, and colleagues at UCLA, establishing the paradigm of leveraging citizens' mobile devices as sensor networks[2]. In 2011, Ganti et al. formally defined the term "Mobile Crowdsensing (MCS)," and thousands of papers have since been published, primarily in IEEE and ACM venues. Capponi et al.'s 2019 survey documented over 300 MCS papers published between 2019 and 2023 alone.

2.2 Volunteered Geographic Information (VGI)

In 2007, Goodchild at UC Santa Barbara proposed the concept of "Citizens as Sensors," establishing the theoretical foundation for citizen-generated geographic data exemplified by OpenStreetMap[3]. The related field of Public Participation GIS (PPGIS), conceptualized in 1996, has evolved into commercial platforms such as Finland's Maptionnaire.

2.3 Emotional Cartography and Urban Emotions

In 2009, Christian Nold established the field through his "Bio Mapping" project, equipping over 2,000 participants across 25+ cities with GSR sensors and GPS to map emotions during walking[4]. Between 2014 and 2020, Resch at the University of Salzburg and Zeile at TU Kaiserslautern led the "Urban Emotions" initiative, integrating wearable biosensors, the "People as Sensors" smartphone app, and social media emotion extraction[5]. This DFG-funded interdisciplinary project positioned citizens' emotional responses as a new data layer for urban planning decision support.

2.4 NLP and Sentiment Analysis in Urban Planning

According to Fu's (2024) review in the Journal of the American Planning Association, an analysis of 55 papers in planning journals revealed that topic modeling (27%), sentiment analysis (19%), and Word2Vec (17%) were the dominant techniques[6]. Between 2024 and 2026, LLM integration has progressed rapidly, with BERT-based sentiment estimation combined with computer vision being applied as "Urban Sentiment Maps" in urban planning practice[7].

The critical point is that these fields remain mutually fragmented. Participatory sensing specializes

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[2]Burke, J., Estrin, D., et al. (2006). Participatory Sensing. ACM SenSys 2006 Workshop on World-Sensor-Web.

[3]Goodchild, M. F. (2007). Citizens as sensors: the world of volunteered geography. GeoJournal, 69(4), 211-221.

[4]Nold, C. (2009). Emotional Cartography: Technologies of the Self. Published by the author. Bio Mapping project conducted across 25+ cities with 2,000+ participants.

[5]Resch, B., Summa, A., Sagl, G., Zeile, P., & Exner, J.-P. (2015). Urban Emotions — Geo-Semantic Emotion Extraction from Technical Sensors, Human Sensors and Crowdsourced Data. In Progress in Location-Based Services 2014, Springer.

[6]Fu, X. (2024). Natural Language Processing in Urban Planning: A Research Agenda. Journal of the American Planning Association.

[7]Jiang, Y., et al. (2025). Exploring the construction and application of "urban sentiment maps" in urban planning practice. Environment and Planning B.

in quantitative sensor data; emotional cartography remains limited to small-scale artistic experiments; NLP research targets short-form social media text. An integrative framework that processes walking-collected narrative data with AI for urban infrastructure management remains unestablished.

3. Representative Peer-Reviewed Publications

3.1 Foundational Papers in Participatory Sensing

Lane, Miluzzo et al.'s (2010) survey in IEEE Communications Magazine recorded over 5,000 citations, systematizing sensing architectures across individual, group, and community scales. Guo et al. (2015) established a comprehensive taxonomy of MCSC (Mobile Crowd Sensing and Computing) in ACM Computing Surveys, distinguishing participatory, opportunistic, and hybrid sensing modes.

3.2 Emotion and Narrative Studies

Quercia, Schifanella, and Aiello (2014) presented "The Shortest Path to Happiness" at ACM Hypertext, constructing an algorithm that recommends emotionally pleasant walking routes based on crowdsourced subjective perceptions (beauty, quietness, happiness) from over 3,300 contributors[8]. They demonstrated that adding just a few minutes to the shortest route improved beauty/happiness scores by 26–30%.

Paiva et al. (2023) published in The Professional Geographer, demonstrating a methodology in which 19 participants wearing EDA sensors during walking in a Lisbon high street were presented with their own biodata[9]. This "bounded interference" was shown to elicit deep narrative reflections about spatial experience. The EU Horizon 2020 "eMOTIONAL Cities" project (2021–2025, €5 million budget) similarly conducted neuroscience-urban planning integration experiments across Lisbon, Copenhagen, London, and East Lansing[10].

3.3 Walking Methods and Crowdsourcing

Evans and Jones (2011) established the "Walking Interview" methodology in Applied Geography, demonstrating that walking interviews generate significantly more place-linked data than seated interviews[11]. They established the "spatial transcript" technique integrating GPS tracking with qualitative GIS. Saha et al.'s (2019) Project Sidewalk at ACM CHI expanded crowdsourced sidewalk accessibility data collection via virtual walking on Google Street View to over 35 cities and 1.1 million labels[12].

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[8] Quercia, D., Schifanella, R., & Aiello, L. M. (2014). The Shortest Path to Happiness. ACM Hypertext 2014.

[9] Paiva, D., et al. (2023). Communicating the Urban Experience through Biosensing: A Participatory Approach. The Professional Geographer, 75(6).

[10] eMOTIONAL Cities (2021-2025). EU Horizon 2020 Project ID 945307. https://emotionalcities-h2020.eu/

[11] Evans, J., & Jones, P. (2011). The walking interview: Methodology, mobility and place. Applied Geography, 31(2), 849-858.

[12] Saha, M., et al. (2019). Project Sidewalk: A Web-based Crowdsourcing Tool for Collecting Sidewalk Accessibility Data at Scale. ACM CHI 2019.

4. Existing Products and Services

Numerous products and services exist for citizen-participatory urban data collection, but none integrates all four elements of HOHOHO (walking × narrative/emotion × AI × urban infrastructure). The following table presents representative examples with a four-dimensional evaluation.

Project	Citizen Participation	Walking Perspective	Narrative/ Emotion	AI Integration	Overall
Walk21 App	★★★★★	★★★★★	★★★★	★★	4/5
Emotional Maps (CZ)	★★★★★	★★★★	★★★★★	★★	4/5
GoodCityLife	★★★	★★★★	★★★★★	★★★★	4/5
Project Sidewalk	★★★★★	★★★★	★	★★★★★	3/5
Mapillary (Meta)	★★★★★	★★★★★	★	★★★★★	3/5
SeeClickFix	★★★★★	★★★	★★	★★	2/5
HOHOHO (proposed)	★★★★★	★★★★★	★★★★★	★★★★★	**5/5**

Table 1: Four-dimensional comparison of HOHOHO with existing models. ★ indicates 5-point scale ratings.

A notable case is Boston's StreetBump application, which used vehicle accelerometers to auto-detect potholes. The app generated a "sample bias" where data concentrated in wealthier neighborhoods due to differential smartphone adoption rates[13]. This lesson validates HOHOHO's design philosophy of intentionally deploying walkers to school routes and historically underserved areas.

[13]Crawford, K. (2013). The Hidden Biases in Big Data. Harvard Business Review, April 2013. — StreetBump bias documented by FTC (2016).

5. Structural Gaps in Existing Research

This survey identifies six structural gaps. These represent findings where both the Claude and Gemini surveys converged.

Gap 1: Disconnect between qualitative narrative data and AI/ML processing. Rich qualitative narrative data from walking interviews and psychogeography field notes have established collection methods, but applying NLP/LLMs to automatically generate urban planning insights from these data is virtually nonexistent. Evans and Jones's (2011) "spatial transcript" technique remains dependent on manual analysis, with no AI processing pipeline established.

Gap 2: Absence of multimodal narrative-emotion-spatial integration systems. No analytical framework exists that integrates text narratives, biosensor data, spatial context, and visual/acoustic data. As the Nature (2025) Urban Visual-Spatial Intelligence paper notes, the lack of robust methodologies for effectively integrating qualitative human perception data with quantitative sensor data remains the foremost challenge[14].

Gap 3: No established protocol for walking-based data collection. A standardized framework for leveraging the multimodal data of walking citizens—GPS, accelerometer, subjective reports, biosensors, and photographs—as systematic urban planning input has not been established.

Gap 4: Lack of longitudinal emotional urban monitoring. All existing studies are cross-sectional snapshots. No system tracks how citizens' narrative and emotional relationships to places change over time.

Gap 5: Absence of sensing-to-planning feedback loops. Closed-loop systems connecting citizen sensing → data analysis → urban planning decisions → implementation → citizen sensing feedback are rare; most remain unidirectional data collection.

Gap 6: Blank space in AI-based infrastructure anomaly detection from citizen narratives. While smartphone accelerometer-based road surface anomaly detection exists, and Project Sidewalk provides visual labeling, the approach of AI automatically extracting infrastructure issues from citizens' walking experience narratives ("this road is dangerous at night," "this sidewalk has terrible steps") has not been researched.

[14]Urban visual-spatial intelligence: linking human and sensor perception for sustainable urban development. npj Urban Sustainability (2025).

6. Potential Originality of the HOHOHO Model

Based on this survey, we identify four dimensions of originality for the HOHOHO model.

Originality 1: Integration model across fragmented research fields

HOHOHO's strongest claim to originality lies in its cross-cutting integration of elements from six or more fields: participatory sensing (computer science), emotional cartography (art/geography), walking interviews (qualitative social science), NLP/AI (data science), citizen science (environmental studies), and urban infrastructure management (civil engineering). No attempt to integrate these into a single walking survey model has been identified in the literature.

Originality 2: Narrative-to-infrastructure AI pipeline

The rapid advancement of LLM technology in 2024–2026 has made it technically feasible for the first time to process citizen narrative data through NLP/LLMs and automatically classify infrastructure issues with spatial mapping. If HOHOHO can construct an end-to-end pipeline of "citizen walking narrative → LLM processing → infrastructure anomaly extraction → spatial analysis → planning recommendations," this would constitute academically uncharted territory.

Originality 3: A new data category of "qualitative citizen intelligence"

Existing urban data falls into three categories: quantitative IoT sensor data, passive VGI from social media mining, and structured reports from civic reporting apps. HOHOHO proposes a fourth category: "unstructured narrative, emotional, and visual data actively generated by walking citizens." This captures a subjective, experiential layer invisible to Google Street View while containing far richer contextual information than SeeClickFix-type structured reports.

Originality 4: Contextual advantage as a Japan-originated model

Japan's super-aging society, pedestrian-focused infrastructure monitoring needs, and high-density urban environments requiring fine-grained quality-of-life visualization present unique conditions. As a model addressing these Japan-specific needs, HOHOHO holds potential for distinctive international contributions.

7. Results and Limitations of LLM Cross-Validation

We report findings from the methodological feature of this survey—the multi-LLM cross-validation. A systematic hallucination check was performed on Gemini DeepResearch output by Claude, yielding the following insights.

7.1 Convergent Findings

The principal findings where both LLMs converged are: (1) Each constituent element of HOHOHO is rooted in established fields, but the integrated model has no precedent. (2) HOHOHO is positioned at the intersection of participatory sensing, emotional cartography, and NLP urban analysis. (3) It represents a structurally different paradigm from FixMyStreet/SeeClickFix-type platforms.

7.2 Issues Detected in Gemini Output

Of Gemini's 54 citations, 8–10 pointed to sources unrelated to the claims they supported. The most severe instance was a legal case analysis on Iowa sidewalk liability whose citation pointed to a small municipality's city council meeting agenda PDF. The StreetBump bias analysis cited an employment discrimination law paper. Additionally, the Iowa Supreme Court case date was off by one year (stated 2023; actual ruling was June 14, 2024)[15].

Furthermore, terms such as "the Liability Paradox" and "Judgment as a Service (JaaS)" were presented as established academic concepts, but these appear to be neologisms coined by Gemini within the report. Established usage in existing literature could not be confirmed. The present paper does not adopt these terms.

7.3 Methodological Implications

This survey's experience empirically validates the premise that "DeepResearch first drafts invariably contain hallucinations." The mechanical assignment of citation URLs is a tendency worth documenting as a Gemini-specific pattern. Multi-LLM cross-validation is suggested as an effective technique for mitigating the risks inherent in single-LLM dependency for academic surveys.

[15]Splittgerber v. Bankers Trust Co., Iowa Supreme Court, June 14, 2024. — Overruled Madden v. City of Iowa City (2014).

8. Conclusion and Future Directions

The conclusion of this survey is unambiguous. While HOHOHO's constituent elements are each rooted in established research fields, their combination is unprecedented in both academic and practical terms. No end-to-end model that "actively collects narrative and emotional data from walking citizens, processes it with AI/LLMs to detect infrastructure anomalies, and integrates the results into urban planning" exists in the world's peer-reviewed literature or commercial product landscape.

As a future research strategy, HOHOHO should establish its originality through dialogue with the following academic communities: ACM UbiComp/CHI (participatory sensing and HCI), AGILE/GIScience (spatial information and VGI), and AESOP/ACSP (urban planning). Explicit comparison and differentiation with the research groups of Resch (University of Salzburg) on Urban Emotions, Quercia (Nokia Bell Labs) on GoodCityLife, and Froehlich (University of Washington) on Project Sidewalk would most effectively demonstrate HOHOHO's novelty.

From the perspective of cross-sector collaboration, the six gaps identified in this survey directly constitute a joint research agenda. Gaps 1, 3, and 6 can be addressed by HOHOHO's field operations, while Gaps 2, 4, and 5 can be filled by an AI technology partner's LLM/Vision capabilities. This complementary relationship provides the structural foundation for industry-academia-government collaboration.

Japan's National Resilience Implementation Mid-Term Plan (FY2026–2030), with its allocation exceeding 20 trillion yen, represents an optimal entry window. The proposition of "walking to digitize the places vehicle-mounted AI cannot reach" offers a clear value proposition for capturing the interstices of this vast market, while simultaneously contributing to the international academic frontier of human-centric urban sensing.

Disclaimer: *This survey incorporates analysis by LLMs (Claude / Gemini). Please consult original publications for details of cited works. While hallucination checks have been performed, completeness is not guaranteed.*